

TrES-3b

R-2824

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210810 · created 2026-09-05T22:50:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459436.7827 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.055
Transit depth (Rp/Rs)^2	1.16 +/- 1.18 [%]
Orbital Inclination (inc)	86.79 +/- 2.95
Airmass coefficient 1 (a1)	0.987 +/- 0.057
Airmass coefficient 2 (a2)	0.015 +/- 0.072
Scatter in the residuals of the lightcurve fit is	5.75 %
Best Comparison Star	#2 - [356, 295]
Optimal Aperture	2.68
Optimal Annulus	5.94
Transit Duration (day)	0.072 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.108 $\pm$ 0.055 vs 0.16309 (deviation 1.0 σ)
Duration fit vs published	0.072 d vs 0.0567 d (deviation 1.18 σ)
Mid-time O–C	3.1 min
Depth SNR	2.0
Residual scatter	5.75 %

Light curve

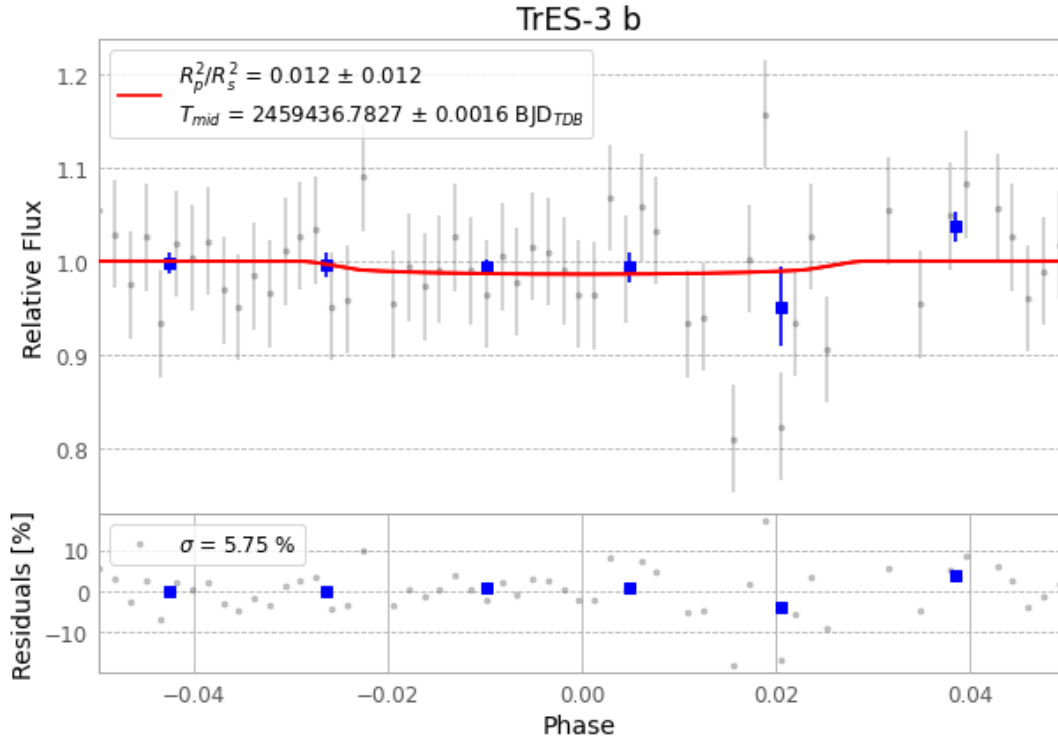


Figure 1 — FinalLightCurve_TrES-3 b_2021-08-09.png (SHA-256 20cf9aa043956308...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [426, 211], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b7cc8f8422fae6ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-3210810050912.FITS → TRES-3210810081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210810.log (a0effe7ea706...), FinalLightCurve_TrES-3 b_2021-08-09.png (20cf9aa04395...), FinalLightCurve_TrES-3 b_2021-08-09.csv (57abbe05f3e8...)

Compute resources

Wall time 227.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 22:50Z.